

# M-Theory on Calabi–Yau Threefolds, Brane Algebras, and Koszul Duality

Physics Notes for Volume III

Raeez Lorgat

April 2026

## Contents

These notes spell out the physical picture behind the quantum vertex chiral group  $G(X)$  of a Calabi–Yau threefold  $X$ . The four ingredients — M-theory holography, open string field theory from CY categories, closed strings from cyclic homology, and the packaging of both into a single algebraic object via Koszul duality — are developed in turn. Throughout, we connect to the rigorous constructions of the monograph: the CY-to-chiral functor  $\Phi$  (Theorem CY-A), the bar-cobar machine of Volume I, and the  $E_1/E_2$  chiral hierarchy of the present Volume III.

## 1 M-theory setup and the two algebras

### 1.1 The geometry

Fix a compact (or suitably non-compact) Calabi–Yau threefold  $X$ . Consider M-theory on the eleven-dimensional background

$$\mathbb{R}_t \times X \times \mathbb{R}^4,$$

where  $\mathbb{R}_t$  is the “time” direction along which operator products will be taken. Place a stack of  $N$  M2-branes supported on

$$\mathbb{R}_t \times \{p\} \subset \mathbb{R}_t \times X,$$

where  $p \in X$  is a point. The M2-branes are thus (1+0)-dimensional objects (quantum-mechanical from the brane worldvolume perspective) wrapping the time direction and localized at a point in  $X$ .

*Remark 1.1.* The choice of  $\mathbb{R}^4$  as the transverse factor is dictated by the requirement that the M2-brane worldvolume theory be topological along  $X$  and that the bulk gravitational theory reduce to a tractable form after compactification on  $X$ . In the topological string setting, one replaces  $\mathbb{R}^4$  by  $\mathbb{C}^{0|2}$  (the odd cotangent directions of the Calabi–Yau). This replacement is the AKSZ origin of the BV formalism on the brane.

## 1.2 The brane algebra $\mathcal{A}_N$

The stack of  $N$  M2-branes supported on  $\mathbb{R}_t \times \{p\}$  carries a worldvolume theory. The algebra of local operators of this quantum-mechanical system is the *brane algebra*

$$\mathcal{A}_N = \text{algebra of operators on the } N\text{-brane stack at } p \in X.$$

The operator product expansion (OPE) in the  $\mathbb{R}_t$  direction equips  $\mathcal{A}_N$  with the structure of an associative algebra. More precisely,  $\mathcal{A}_N$  is a differential graded (dg) algebra in the BV formalism, with  $Q$ -cohomology the physical operator algebra.

*Remark 1.2* (Explicit form for  $X = \mathbb{C}^3$ ). When  $X = \mathbb{C}^3$  (or more generally a toric CY3), the brane worldvolume theory is a matrix quantum mechanics. In the BV formalism, the fields are

$$\alpha \in \Omega^*(\mathbb{R}_t)[\varepsilon_1, \varepsilon_2, \varepsilon_3] \otimes \mathfrak{gl}_N[1],$$

where the  $\varepsilon_i$  are odd variables representing the three normal directions in  $X$  (the Ext algebra  $\text{Ext}_{\mathcal{O}_X}^*(\mathcal{O}_p, \mathcal{O}_p) = \bigwedge(\varepsilon_1, \varepsilon_2, \varepsilon_3)$  for a point  $p \in \mathbb{C}^3$ ). The action is

$$S(\alpha) = \int_{\mathbb{R}_t \times \mathbb{C}^{0|3}} \left( \frac{1}{2} \text{Tr}(\alpha d\alpha) + \frac{1}{3} \text{Tr}(\alpha^3) \right),$$

where the integral over  $\mathbb{C}^{0|3}$  is the Berezin integral over the odd fiber. De-BV-fying (extracting the ghost-number-zero fields), one obtains a gauge field  $A \in \Omega^1(\mathbb{R}_t) \otimes \mathfrak{gl}_N$  and three  $\mathfrak{gl}_N$ -valued scalars  $\phi_1, \phi_2, \phi_3$  with action

$$S = \int_{\mathbb{R}_t} \text{Tr}(\phi_1 d_A \phi_2 + \phi_3 [\phi_1, \phi_2]).$$

This is the dimensional reduction of holomorphic Chern–Simons theory on  $\mathbb{C}^3$  to  $\mathbb{R}_t$ , i.e., a matrix model with cubic potential  $W = \text{Tr}(XYZ - XZY)$ .

## 1.3 The gravitational algebra $\mathcal{B}$

The *gravitational algebra*  $\mathcal{B}$  is the algebra of local operators of the topological/holomorphic gravitational theory on the full spacetime  $\mathbb{R}_t \times X$ . Concretely, after compactification on  $X$ , this is the BCOV-type theory whose fields are polyvector fields on  $X$  (subject to the  $\partial$ -closedness condition), and whose OPE in  $\mathbb{R}_t$  provides the associative product.

For  $X = \mathbb{C}^3$  (or a neighborhood thereof), the gravitational theory is the Kodaira–Spencer theory:

$$\begin{aligned} \text{Fields: } \quad & \alpha \in \Omega^*(\mathbb{R}_t) \hat{\otimes} \Omega^{0,*}(X)[1], \\ & \beta \in \Omega^*(\mathbb{R}_t) \hat{\otimes} \Omega^{0,*}(X)[2], \\ \text{Action: } \quad & S = \int_{\mathbb{R}_t \times X} \beta \bar{\partial} \alpha \Omega + \beta \partial \alpha \wedge \partial \alpha, \end{aligned} \tag{1}$$

where  $\Omega$  is the holomorphic volume form. The second term encodes the Poisson bracket on Hamiltonian vector fields:  $\partial \alpha \wedge \partial \alpha = \{\alpha, \alpha\} \Omega$ .

The ring of classical local operators is

$$\mathcal{B}^{\text{cl}} = C^*(\mathbb{C}[z_1, z_2, z_3] \ltimes \mathbb{C}[z_1, z_2, z_3][1]),$$

the Chevalley–Eilenberg cochains of the semi-direct product of the polynomial ring (viewed as a Lie algebra via the Poisson bracket  $\{-, -\}$  from  $\Omega$ ) with itself shifted, viewed as the adjoint module.

## 1.4 Holography as Koszul duality

The holographic duality between the brane theory and the gravitational theory admits a sharp algebraic formulation:

**Conjecture 1.3** (Holographic Koszul duality). *In the large- $N$  limit, the brane algebra and the gravitational algebra are related by Koszul duality:*

$$\lim_{N \rightarrow \infty} \mathcal{A}_N = \mathcal{B}^!,$$

where  $\mathcal{B}^!$  denotes the Koszul dual of  $\mathcal{B}$ .

*Remark 1.4* (Evidence in the  $\mathbb{C}^2$  example). The original notes verify this for  $X = \mathbb{C}^2$  (mixed A-B model on  $\mathbb{R}_A^2 \times \mathbb{C}_B^2$ ). There, the gauge theory operators are  $\text{Sym}^*(\mathbb{C}[z_1, z_2] \oplus \mathbb{C}[z_1, z_2][1])$  and the gravity operators are  $C^*(\mathbb{C}[z_1, z_2] \ltimes \mathbb{C}[z_1, z_2][1])$ . The Koszul dual of the latter is  $\mathbf{U}(\text{Hom}(\mathbb{C}^2) \oplus \text{Hom}(\mathbb{C}^2)[1])$ , whose commutative (classical, large- $N$ ) limit is the former. Koszul duality thus interchanges:

|                                |                       |                                        |
|--------------------------------|-----------------------|----------------------------------------|
| Classical free gravity         | $\longleftrightarrow$ | Classical gauge theory                 |
| Poisson bracket (interactions) | $\longleftrightarrow$ | First quantum correction ( $\hbar^1$ ) |
| Genus- $g$ string amplitude    | $\longleftrightarrow$ | $\hbar^g$ correction (ribbon graphs)   |

The  $A_\infty$ -operations in the gravitational Chevalley–Eilenberg complex (Feynman diagrams with  $m_2, m_3, \dots$ ) are matched by multi-trace operators and their OPE on the gauge theory side.

*Remark 1.5* (Nature of the limit). The “ $\lim_{N \rightarrow \infty}$ ” should be understood as follows. For each finite  $N$ , the brane algebra  $\mathcal{A}_N$  involves  $\mathfrak{gl}_N$ -valued fields. Invariant theory provides the algebra of gauge-invariant operators

$$\mathcal{A}_N^{\text{inv}} = \{\text{Tr}(P(\phi_1, \dots, \phi_d)) : P \text{ non-commutative polynomial}\} / \text{trace relations at rank } N.$$

As  $N \rightarrow \infty$  the trace relations disappear and the generators become free:  $\mathcal{A}_\infty^{\text{inv}}$  is the free algebra on single-trace generators. This free algebra is Koszul dual to the Chevalley–Eilenberg cochains of the gravitational Lie algebra.

## 2 Open strings from CY categories

### 2.1 The CY category as the category of branes

The data of M-theory on  $\mathbb{R}_t \times X$  includes a category of branes:

- In the B-model, branes are coherent sheaves on  $X$ :  $\mathcal{C} = D^b(\text{Coh}(X))$ .
- In the A-model, branes are Lagrangians:  $\mathcal{C} = \text{Fuk}(X)$ .
- In mixed models (e.g.,  $X = \mathbb{R}_A^2 \times \mathbb{C}_B^2$ ), branes are products  $L \times Z$  of Lagrangians and coherent sheaves.

In every case,  $\mathcal{C}$  is a *Calabi–Yau category of dimension  $d = \dim_{\mathbb{C}} X$* : it carries a cyclic  $A_{\infty}$ -structure induced by the holomorphic volume form  $\Omega \in H^0(X, K_X)$ .

## 2.2 Open string fields as $\text{RHom}$

Fix a brane  $\mathcal{F} \in \mathcal{C}$  (for instance, the skyscraper sheaf  $\mathcal{O}_p$  for M2-branes at a point  $p$ , or a Lagrangian  $L$  for an A-brane). The space of *open string fields* is

$$\boxed{\text{Open string fields} = \text{RHom}_{\mathcal{C}}(\mathcal{F}, \mathcal{F})[1].}$$

The shift  $[1]$  is the BV shift: it places the connection (gauge field) in ghost number zero and the matter fields (normal deformations of the brane) in appropriate ghost numbers.

**Example 2.1** (Point brane on  $\mathbb{C}^3$ ). For  $\mathcal{F} = \mathcal{O}_p$  with  $p \in \mathbb{C}^3$ , the self-Ext algebra is

$$\text{Ext}_{\mathcal{O}_{\mathbb{C}^3}}^*(\mathcal{O}_p, \mathcal{O}_p) = \bigwedge(\varepsilon_1, \varepsilon_2, \varepsilon_3),$$

an exterior algebra on three generators of degree 1 (one per normal direction). After the BV shift and tensoring with  $\mathfrak{gl}_N$  for  $N$  branes, this gives the field content of Remark ??.

**Example 2.2** (Point brane on  $\mathbb{C}^2$ ). For the mixed A-B model on  $\mathbb{R}^2 \times \mathbb{C}^2$ , the brane  $\mathcal{F} = \mathbb{R} \times \{p\}$  has open string fields

$$\Omega^*(\mathbb{R}) \otimes \text{Ext}_{\mathcal{O}_{\mathbb{C}^2}}^*(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1] = \Omega^*(\mathbb{R})[\varepsilon_1, \varepsilon_2] \otimes \mathfrak{gl}_N[1],$$

the Floer–Ext tensor product.

## 2.3 The open string field theory action from $A_{\infty}$ -structure

The CY category  $\mathcal{C}$  carries an  $A_{\infty}$ -structure: higher composition maps  $\mu_n: \text{Hom}(X_0, X_1) \otimes \cdots \otimes \text{Hom}(X_{n-1}, X_n) \rightarrow \text{Hom}(X_0, X_n)[2-n]$ . For the endomorphism algebra  $A = \text{End}_{\mathcal{C}}(\mathcal{F})$  of a single brane, the  $A_{\infty}$ -operations  $\mu_1, \mu_2, \mu_3, \dots$  determine the open string field theory action by the Witten–Gaberdiel–Zwiebach prescription:

$$S(\alpha) = \sum_{n \geq 1} \frac{1}{n+1} \langle \alpha, \mu_n(\alpha, \dots, \alpha) \rangle,$$

where  $\langle \cdot, \cdot \rangle: A \otimes A \rightarrow k[-d]$  is the CY pairing and  $\alpha \in A[1]$  is the string field.

*Remark 2.3* (Relation to Theorem CY-A). The passage

$$\text{cyclic } A_{\infty}\text{-algebra } (A, \langle \cdot, \cdot \rangle) \longmapsto \text{chiral algebra } \Phi(\mathcal{C})$$

of Theorem CY-A is *precisely* the open string field theory construction in categorical language. The cyclic  $A_{\infty}$ -structure on  $A = \text{End}_{\mathcal{C}}(\mathcal{F})$  is the chain-level datum encoding the full tree-level open string field theory. The CY-to-chiral functor  $\Phi$  promotes this to a factorization algebra on a curve  $C$  by:

- (i) Extracting the Lie conformal algebra  $\mathfrak{L}_{\mathcal{C}}$  from the Gerstenhaber bracket on  $\mathrm{HH}^*(\mathcal{C})$ ;
- (ii) Taking the factorization envelope  $\mathrm{Fact}_{\mathcal{C}}(\mathfrak{L}_{\mathcal{C}})$ ;
- (iii) Enhancing to  $E_2$  via the  $\mathbb{S}^d$ -framing (for  $d = 3$ , the  $\mathbb{S}^3$ -framing gives  $E_3$ , which restricts to  $E_2$  via Dunn additivity);
- (iv) Quantizing using the CY trace.

The upshot: the chiral algebra  $\Phi(\mathcal{C})$  is the open string field theory *organized as a factorization algebra*.

## 2.4 $N$ branes and matrix enlargement

For a stack of  $N$  branes, one replaces the endomorphism algebra by its matrix enlargement:

$$A_N = \mathrm{End}_{\mathcal{C}}(\mathcal{F}^{\oplus N}) \simeq \mathrm{End}_{\mathcal{C}}(\mathcal{F}) \otimes \mathfrak{gl}_N.$$

The  $A_{\infty}$ -structure extends by the Leibniz rule on the  $\mathfrak{gl}_N$  factor. The brane algebra  $\mathcal{A}_N$  is the  $Q$ -cohomology of the resulting dg algebra of gauge-invariant operators:

$$\mathcal{A}_N = H^*\left(\left(\mathrm{Sym}^*(A_N[1])\right)^{\mathfrak{gl}_N}, Q_{\mathrm{BV}}\right).$$

Concretely, generators are single-trace operators  $\mathrm{Tr}(P(\phi_1, \dots, \phi_d))$  with  $P$  a non-commutative polynomial in the matter fields (and ghosts/anti-fields at other ghost numbers). As  $N \rightarrow \infty$ , the trace relations vanish and  $\mathcal{A}_{\infty}$  is freely generated by single traces.

## 3 Closed strings from cyclic homology

### 3.1 The theorem of Loday–Quillen–Tsygan

The passage from open strings to closed strings is controlled by a classical theorem in non-commutative geometry:

**Theorem 3.1** (Loday–Quillen–Tsygan). *For a unital associative algebra  $A$ , the Chevalley–Eilenberg homology of the Lie algebra of infinite matrices over  $A$  is an exterior algebra generated by the cyclic homology of  $A$ :*

$$H_*^{\mathrm{CE}}(\mathfrak{gl}_{\infty}(A)) \simeq \bigwedge \mathrm{HC}_*(A).$$

### 3.2 Physical interpretation

The physical content of Loday–Quillen–Tsygan is as follows:

- (1) **The open string algebra** is  $A = \mathrm{End}_{\mathcal{C}}(\mathcal{F})$ , the endomorphism algebra of a brane.
- (2) **Matrix enlargement and the large- $N$  limit.** The Lie algebra  $\mathfrak{gl}_N(A) = \mathfrak{gl}_N \otimes A$  is the open string algebra for  $N$  branes. Taking  $N \rightarrow \infty$  yields  $\mathfrak{gl}_{\infty}(A)$ .

- (3) **Chevalley–Eilenberg homology = gauge-invariant operators.** The CE chain complex  $C_*^{\text{CE}}(\mathfrak{gl}_\infty(A))$  computes the homology of gauge-invariant observables in the large- $N$  matrix model. This is the space of *closed string states*: the gauge-invariant sector of the open string theory IS the closed string theory.
- (4) **Single-trace generators = single-string states.** The exterior algebra generators  $\text{HC}_*(A)$  correspond to single-trace operators  $\text{Tr}(\phi_{i_1} \cdots \phi_{i_k})$ . These are the *single-string states*: each cyclic homology class is one closed string mode.
- (5) **Multi-trace operators = multi-string states.** The exterior power  $\bigwedge^k \text{HC}_*(A)$  corresponds to  $k$ -trace operators, i.e.,  $k$ -string states. The exterior algebra structure encodes the statistics (closed strings are bosonic/fermionic depending on degree).
- (6) **OPE = string scattering.** The OPE of single-trace operators on the gauge theory side matches the tree-level closed string scattering amplitudes. Multi-trace OPE (with trace-splitting and trace-joining) matches higher-genus string amplitudes.

### 3.3 The closed string algebra as chiral derived center

In the language of the monograph:

**Proposition 3.2** (Closed strings = derived center). *The closed string algebra produced by Loday–Quillen–Tsygan is the chiral derived center  $Z_{\text{ch}}^{\text{der}}(\Phi(\mathcal{C}))$  of the chiral algebra associated to the CY category:*

$$\text{Closed string algebra} = H_*^{\text{CE}}(\mathfrak{gl}_\infty(A)) \simeq \bigwedge \text{HC}_*(A) = Z_{\text{ch}}^{\text{der}}(\Phi(\mathcal{C})).$$

The identifications proceed as follows:

- $\text{HC}_*(A)$  is the cyclic homology of the open string algebra — equivalently, the Hochschild homology of  $\mathcal{C}$  with its  $S^1$ -action.
- $\text{HC}_*(A)$  is simultaneously the space of deformations of the closed string background (via the Kontsevich–Soibelman map  $\text{HC}_*(A) \rightarrow \text{deformations of the } A_\infty\text{-structure}$ ).
- The chiral derived center  $Z_{\text{ch}}^{\text{der}}(\Phi(\mathcal{C}))$  is the  $E_2$ -center of the chiral algebra — exactly the algebra of bulk (gravitational) local operators.

*Remark 3.3.* The identification of the derived center with the gravitational algebra is the categorical incarnation of the Drinfeld center construction:

$$\mathcal{Z}(\text{Rep}^{E_1}(\Phi(\mathcal{C}))) \simeq \text{Rep}^{E_2}(Z_{\text{ch}}^{\text{der}}(\Phi(\mathcal{C}))).$$

The monoidal  $(E_1)$  representation category of the boundary (brane) theory determines a braided monoidal  $(E_2)$  category via the center construction. This is the bulk-boundary correspondence at the categorical level.

## 4 The gravitational algebra from polyvector fields

### 4.1 BCOV theory on $X$

The gravitational (closed string field) theory on  $\mathbb{R}_t \times X$  is the Kodaira–Spencer / BCOV theory. Following the analysis in the earlier notes, and extending from  $\mathbb{C}^2$  to a general CY threefold  $X$ :

The space of closed string states (before interaction) is

$$\ker \partial \subset \mathbf{PV}^{*,*}(X) \simeq \bigoplus_{p,q} \Omega^{0,q}(X, \bigwedge^p TX),$$

where  $\partial: \mathbf{PV}^{p,q} \rightarrow \mathbf{PV}^{p-1,q}$  is the divergence operator (contraction with  $\Omega$ ). The components:

- $\mathbf{PV}^{0,*}(X) \subset \ker \partial$ : functions (automatic).
- $\mathbf{PV}^{1,*}(X) \cap \ker \partial$ : divergence-free vector fields, i.e., Hamiltonian vector fields (via  $\Omega$ , these are identified with closed holomorphic  $(d-1)$ -forms).
- $\mathbf{PV}^{2,*}(X) \cap \ker \partial$ :  $\partial$ -closed bi-vector fields. For  $X = \mathbb{C}^3$ , these give deformations of the holomorphic symplectic structure.
- $\mathbf{PV}^{3,*}(X) \cap \ker \partial$ : the “top” polyvector fields, corresponding to deformations of  $\Omega$  itself.

### 4.2 The gravitational algebra as Chevalley–Eilenberg cochains

After the reduction to physical operators (passing to  $Q$ -cohomology), the ring of classical local operators of the gravitational theory at a point  $x \in X$  is

$$\mathcal{B}^{\text{cl}} = C^*(\mathfrak{gl}_\Omega(X) \ltimes \mathfrak{gl}_\Omega(X)[1]),$$

where  $\mathfrak{gl}_\Omega(X) = \{f \in \mathcal{O}(X) : \{f, -\} \text{ preserves } \Omega\}$  is the Lie algebra of Hamiltonian functions with Poisson bracket  $\{-, -\}$  determined by  $\Omega$ . The semi-direct product structure reflects the BF-type nature of the theory:  $\alpha$ -fields (Hamiltonian vector fields) act on  $\beta$ -fields (adjoint-valued functions) via the Poisson bracket, while  $\beta$  commutes with  $\beta$ .

For computations at a formal neighborhood of  $p \in X \simeq \mathbb{C}^3$ :

$$\begin{aligned} \text{Ghost \#1 operators: } \mathcal{O}_i^{(1)} &= \partial_{z_1}^a \partial_{z_2}^b \partial_{z_3}^c \alpha \quad (a, b, c \geq 0), \\ \text{Ghost \#2 operators: } \mathcal{O}_i^{(2)} &= \partial_{z_1}^a \partial_{z_2}^b \partial_{z_3}^c \beta \quad (a, b, c \geq 0). \end{aligned}$$

These transform *contravariantly* to the gauge theory operators: the gravitational operators carry polynomial derivatives (jet space), while the gauge theory operators carry polynomial values (formal functions). This contravariance is the fingerprint of Koszul duality.

## 5 The Koszul duality dictionary

### 5.1 Classical Koszul duality

At the classical level, the Koszul duality between gauge theory and gravity takes the form:

|                   | <b>Gauge theory (brane)</b>                        | <b>Gravity (bulk)</b>                        |
|-------------------|----------------------------------------------------|----------------------------------------------|
| <b>Algebra</b>    | $\mathcal{A}_\infty = \text{Sym}^*(V \oplus V[1])$ | $\mathcal{B} = C^*(V \ltimes V[1])$          |
| <b>Type</b>       | Commutative (symmetric)                            | Lie (CE cochains)                            |
| <b>Generators</b> | $\text{Tr}(\phi^I)$ (polynomial values)            | $\partial^I \alpha, \partial^I \beta$ (jets) |
| <b>Product</b>    | Commutative product                                | Chevalley–Eilenberg cup product              |

where  $V = \mathbb{C}[z_1, \dots, z_d]$  (polynomial functions on the transverse CY directions). The Koszul dual of  $C^*(V \ltimes V[1])$  is  $\mathbf{U}(V \oplus V[1])$  — the universal enveloping algebra — whose commutative limit (setting  $\hbar \rightarrow 0$ , equivalently  $N \rightarrow \infty$  with classical scaling) is  $\text{Sym}^*(V \oplus V[1])$ .

## 5.2 Quantum corrections: order by order in $\hbar$

Moving beyond the classical limit, Koszul duality becomes a precise matching between quantum corrections:

$\hbar^0$ : Classical free gravity  $\longleftrightarrow$  Classical gauge theory (commutative limit).

$\hbar^1$ : The Poisson bracket  $\{z_1^k z_2^l, z_1^m z_2^n\}$  on the gravity side matches the commutator  $[\text{Tr}(\phi_1^k \phi_2^l), \text{Tr}(\phi_1^m \phi_2^n)]$  on the gauge theory side. Diagrammatically, this is a single propagator connecting two single-trace sub-diagrams.

$\hbar^2$ : By the even quantization symmetry ( $\hbar \mapsto -\hbar$  with reversal of product), there are no corrections at order  $\hbar^2$ .

$\hbar^3$ : The non-trivial  $A_\infty$ -operations appear. On the gauge theory side, ribbon graphs with  $k$  boundaries contribute  $k$ -trace operators. A cubic ribbon graph (three boundaries) gives an  $m_3$  operation; a quadratic one gives a corrected  $m_2$ . On the gravity side, these match Feynman diagrams computing the  $A_\infty$ -structure on  $C^*(\mathfrak{g}_\Omega)$ .

*Remark 5.1* (Genus expansion). The  $\hbar$  expansion is simultaneously a genus expansion. At order  $\hbar^g$ , the diagrams on the gauge theory side are ribbon graphs of genus  $\leq g$ ; on the gravity side, they are genus- $g$  string worldsheets. The precise statement is:

- $\hbar^0$ : tree level (genus 0, planar diagrams)
- $\hbar^1$ : one-loop (torus diagrams, first non-planar correction)
- $\hbar^g$ : genus- $g$  worldsheets  $\longleftrightarrow$  ribbon graphs with  $g$  handles

In the Volume I language, this is the shadow obstruction tower: the genus- $g$  obstruction  $\text{obs}_g$  receives contributions from *all* arities (the genus grading and the arity filtration are independent). The leading genus- $g$  term on the uniform-weight lane is  $\kappa \cdot \lambda_g$ .

## 6 The quantum vertex chiral group $G(X)$

### 6.1 Packaging the two algebras

The quantum vertex chiral group  $G(X)$  associated to a CY3  $X$  is a single algebraic object that packages both the brane algebra and the gravitational algebra, with Koszul duality as the internal organizing principle:

**Definition 6.1** (Quantum vertex chiral group: physics summary). The quantum vertex chiral group  $G(X)$  is the  $E_2$ -chiral algebra  $\Phi(\mathcal{C})$ , where  $\mathcal{C}$  is the CY category of  $X$  (either  $D^b(\text{Coh}(X))$  or  $\text{Fuk}(X)$ ), equipped with the following structure:

- (i) **Positive half (brane algebra).** The  $E_1$ -sector of  $G(X)$  is the CoHA:  $G(X)^+ = \text{CoHA}(\mathcal{C}) \simeq \mathcal{A}_\infty$ . For toric  $X$ , this is  $Y^+(\widehat{\mathfrak{g}}_{Q_X})$  (the positive half of the affine super Yangian).
- (ii) **Derived center (gravitational algebra).** The chiral derived center  $Z_{\text{ch}}^{\text{der}}(G(X)) = \mathcal{B}$  is the gravitational algebra. It is generated as an exterior algebra by  $\text{HC}_*(\mathcal{C})$  (Loday–Quillen–Tsygan).
- (iii) **Koszul duality = holography.** The positive half and the derived center are related by Koszul duality:

$$G(X)^+ \simeq (Z_{\text{ch}}^{\text{der}}(G(X)))^!.$$

This is the algebraic incarnation of holographic duality.

- (iv)  **$E_2$ -structure (braiding =  $R$ -matrix).** The  $E_2$ -chiral enhancement provides the braiding on  $\text{Rep}^{E_2}(G(X))$ . The universal  $R$ -matrix  $R(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{G(X)})$  satisfies the quantum Yang–Baxter equation and encodes the quantum group structure.

### 6.2 The triangular decomposition

As a super Lie algebra (or BKM superalgebra in the  $K3 \times E$  setting),  $G(X)$  admits a triangular decomposition:

$$G(X) = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+,$$

where:

- $\mathfrak{n}_+$  = positive half = CoHA = brane algebra  $\mathcal{A}_\infty$ .
- $\mathfrak{h}$  = Cartan subalgebra = lattice  $\Lambda(X) \otimes \mathbb{R}$  extracted from  $K$ -theory of  $X$ .
- $\mathfrak{n}_-$  = negative half = co-brane algebra (Koszul dual generators).

The root space decomposition

$$\mathfrak{n}_+ = \bigoplus_{\alpha \in \Delta_+} \mathfrak{n}_\alpha, \quad \dim \mathfrak{n}_\alpha = \text{mult}(\alpha)$$

is controlled by the BPS spectrum:  $\text{mult}(\alpha) = \text{DT}_\alpha(X)$  (Donaldson–Thomas invariant of class  $\alpha$ ). The real roots correspond to rigid BPS states; the imaginary roots to families.

### 6.3 The automorphic correction and the shadow obstruction tower

The passage from the “tree-level” root datum (real roots + leading imaginary roots) to the full BKM superalgebra is the *automorphic correction*. In the language of Volume I:

| BKM / physics                        | Volume I bar-cobar                                                      |
|--------------------------------------|-------------------------------------------------------------------------|
| Tree-level $r$ -matrix               | Collision $r$ -matrix $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_A)$ |
| Imaginary root corrections           | Higher-arity shadows $(C, Q, \dots)$                                    |
| Full automorphic correction          | Full MC element $\Theta_A$                                              |
| Denominator identity $\Phi(z)$       | Bar-complex Euler product                                               |
| Automorphic form (e.g., $\Delta_5$ ) | Genus- $g$ obstruction $\text{obs}_g$                                   |

The key identification:

$$\text{Automorphic correction} = \text{Shadow obstruction tower}.$$

The shadow obstruction tower  $\Theta_A^{\leq r}$  of Volume I, projected at successive arities, recovers the successive imaginary root corrections to the BKM superalgebra.

### 6.4 The two main examples

#### 6.4.1 Example 1: $X = \mathbb{C}^3$ (affine Yangian)

- Quiver: Jordan quiver (one vertex, one loop with cubic potential  $W = \text{Tr}(XYZ - XZY)$ ).
- $\mathcal{A}_\infty = \text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$  (Schiffmann–Vasserot).
- $G(\mathbb{C}^3) = Y(\widehat{\mathfrak{gl}}_1)$ , the full affine Yangian of  $\mathfrak{gl}_1$ .
- This is isomorphic to  $\mathcal{W}_{1+\infty}$  at the self-dual level — already a central object in Volume I, where MC4 is solved by weight stabilization.
- $\mathcal{B} = \text{Kodaira–Spencer theory on } \mathbb{C}^3 = \text{the chiral derived center}.$
- The Yangian  $R$ -matrix is the DK-0 shadow.

#### 6.4.2 Example 2: $X = (\mathbf{K3} \times E)/(\mathbb{Z}/N\mathbb{Z})$ (BKM superalgebras)

- Lattice:  $\Lambda^{3,2}$  of signature  $(3, 2)$  from  $H^*(X)$ .
- Real roots: three simple roots  $\delta_1, \delta_2, \delta_3$  with Gram matrix  $\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$ .
- Root multiplicities:  $\text{mult}(\alpha) = f(nm, l)$ , where  $f(n, l)$  are Fourier coefficients of the K3 elliptic genus  $\varphi_{0,1}$ .
- $G(X) = \mathfrak{g}_{\Delta_5}$ , the BKM superalgebra whose denominator identity is (up to normalization) the Igusa cusp form  $\Delta_5$ , a Siegel modular form of weight 5 for  $\text{Sp}_4(\mathbb{Z})$ .

- The  $E_2$ -structure is the  $\mathrm{Sp}_4(\mathbb{Z})$  action on the genus-2 Siegel upper half-space  $\mathbb{H}_2$ .
- The shadow obstruction tower captures the automorphic correction: arity 2 is the Weyl vector contribution, higher arities add imaginary roots.

## 7 The full picture: open/closed/holography

We can now assemble the complete dictionary:

| Open strings                                 | $\xrightarrow{\text{LQT}}$ | Closed strings                                       | $\xleftarrow{\text{KD}}$ | Branes          |
|----------------------------------------------|----------------------------|------------------------------------------------------|--------------------------|-----------------|
| $\mathrm{RHom}(\mathcal{F}, \mathcal{F})[1]$ |                            | $\bigwedge \mathrm{HC}_*(\mathrm{End}(\mathcal{F}))$ |                          | $\mathcal{A}_N$ |
| $A_\infty$ -structure                        |                            | $Z_{\mathrm{ch}}^{\mathrm{der}}$                     |                          | CoHA            |
| $E_1$ -sector                                |                            | $E_2$ -bulk                                          |                          | $E_1$ -boundary |

Reading left to right:

- (1) **Open strings.** The CY category  $\mathcal{C}$  provides the space of open string fields  $\mathrm{RHom}(\mathcal{F}, \mathcal{F})[1]$  with action from the  $A_\infty$ -structure. This is Theorem CY-A: the cyclic  $A_\infty$ -structure is promoted to a chiral algebra  $\Phi(\mathcal{C})$ .
- (2) **Closed strings.** Loday–Quillen–Tsygan extracts the closed string algebra as  $\bigwedge \mathrm{HC}_*(\mathrm{End}(\mathcal{F}))$  — the exterior algebra on cyclic homology. This is the chiral derived center  $Z_{\mathrm{ch}}^{\mathrm{der}}(\Phi(\mathcal{C}))$  and equals the gravitational algebra  $\mathcal{B}$ .
- (3) **Holography.** The brane algebra  $\mathcal{A}_\infty$  (single-trace operators, the CoHA, the positive half of  $G(X)$ ) is Koszul dual to  $\mathcal{B}$ . The Drinfeld center construction  $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathcal{A})) \simeq \mathrm{Rep}^{E_2}(\mathcal{B})$  provides the categorical bulk-boundary correspondence.
- (4) **The quantum vertex chiral group.**  $G(X)$  packages all three:  $G(X)^+ = \mathcal{A}_\infty$  (brane algebra / CoHA),  $Z_{\mathrm{ch}}^{\mathrm{der}}(G(X)) = \mathcal{B}$  (gravitational algebra), and Koszul duality between them is holography. The  $E_2$ -structure (braiding /  $R$ -matrix) encodes the quantum group structure, and the full MC element  $\Theta_{G(X)}$  (the automorphic correction) is the shadow obstruction tower capturing all genera.

### 7.1 Summary of identifications

| Physics                 | Algebra                                                                           | Vol I/III                                |
|-------------------------|-----------------------------------------------------------------------------------|------------------------------------------|
| M2-brane operators      | $\mathcal{A}_N$ (gauge-invariant operators)                                       | CoHA, $G(X)^+$                           |
| Gravitational operators | $\mathcal{B}$ (CE cochains)                                                       | $Z_{\text{ch}}^{\text{der}}(G(X))$       |
| Open string fields      | $\text{RHom}(\mathcal{F}, \mathcal{F})[1]$                                        | Cyclic $A_\infty$                        |
| Open string action      | $\sum \frac{1}{n+1} \langle \alpha, \mu_n(\alpha^n) \rangle$                      | CY trace + $A_\infty$                    |
| Closed string states    | $\text{HC}_*(\text{End}(\mathcal{F}))$                                            | Cyclic homology                          |
| Multi-string Fock space | $\bigwedge \text{HC}_*(\text{End}(\mathcal{F}))$                                  | LQT theorem                              |
| Holographic duality     | $\mathcal{A}_\infty = \mathcal{B}^!$                                              | Koszul duality                           |
| Bulk-boundary           | $\mathcal{Z}(\text{Rep}^{E_1}(\mathcal{A})) \simeq \text{Rep}^{E_2}(\mathcal{B})$ | Drinfeld center                          |
| $R$ -matrix             | QYB solution                                                                      | $\text{Res}_{0,2}^{\text{coll}}(\Theta)$ |
| BPS spectrum            | DT invariants                                                                     | Root multiplicities                      |
| String scattering       | OPE of traces                                                                     | $A_\infty$ -operations via KD            |
| Genus- $g$ amplitude    | Ribbon graph, $\hbar^g$                                                           | Shadow obstruction tower arity $g+1$     |

## 8 Closing remarks and open directions

- (1) **Wall-crossing.** The BPS spectrum of  $X$  is subject to wall-crossing (Kontsevich–Soibelman). In the  $G(X)$  language, wall-crossing should correspond to gauge equivalence of the MC element  $\Theta_{G(X)}$ : different stability conditions give gauge-equivalent  $A_\infty$ -structures, hence isomorphic quantum vertex chiral groups. This is the content of the  $A_\infty$ -wall-crossing formula.
- (2) **Beyond CY3.** For CY  $d$ -folds with  $d \neq 3$ , the story changes:
  - $d = 2$  (K3 surfaces): the  $\mathbb{S}^2$ -framing gives an  $E_2$ -algebra directly. No M-theory lift needed; the story is purely about 2d string theory.
  - $d = 4$  (CY 4-folds): the brane is an M5-brane (or the topological twist thereof), and the brane algebra is 3-dimensional. The relevant algebraic structure is an  $E_3$ -algebra (via the  $\mathbb{S}^4$ -framing and Dunn).
- (3) **Non-toric CY3.** For non-toric CY3 (e.g., the quintic), the CoHA construction is more subtle: one lacks a quiver presentation, and the root datum must be extracted from the full derived category or Fukaya category. The conjecture is that  $G(X)$  still exists (as a quantum vertex chiral group) but may not admit a BKM or Yangian presentation.
- (4) **Closed string field theory.** The gravitational algebra  $\mathcal{B}$  is the *classical* closed string field theory. The full quantum closed string field theory is controlled by the genus expansion, i.e., by the full shadow obstruction tower  $\Theta_{G(X)}$ . The quantum master equation  $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$  is the BV master equation of the closed string field theory.

- (5) **Multiple brane types.** With multiple branes  $\mathcal{F}_1, \dots, \mathcal{F}_k$ , the open string algebra is enriched to the full  $A_\infty$ -category  $\mathcal{C}$  (not just an endomorphism algebra). The brane algebra becomes a factorization algebra on the configuration space of brane types. For  $k$  brane types with  $E_2$  directions, one expects  $E_2$ -Koszul duality in the sense of the braided factorization theory of Part IV.